Huy Dinh Tran

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EDUCATION

University of Kansas 08/2023 — 05/2028 (Expected)
Ph.D Computer Science
Advisor: Prof. Mohammad Alian (Cornell University)

University of California, Riverside 09/2021 — 03/2023
M.S. Computer Engineering
Advisor: Prof. Daniel Wong

Pennsylvania State University 08/2017 — 05/2021
B.S. Electrical Engineering

EXPERIENCE

University of Kansas Lawrence, KS
Graduate Research Assistant 08/2023 — Present

- Researched architecture designs to enhance the performance and efficiency of datacenters.
- **Multithreading Architecture — Simultaneous Data-Delivery Threads (SDD):** designed specialized lightweight hardware threads to handle data delivery in server-class workloads that run alongside the main application thread; reduced area by 35% and power by 44%, while maintaining or exceeding throughput and tail latency performance.

Graduate Teaching Assistant 08/2024 — Present

- Taught Software Engineering II (EECS 581) and Computer Science and Interdisciplinary Computing Capstone (EECS 582).
- Supervised 7 senior undergraduate teams on design projects by conducting weekly meetings to provide guidance.

Futurewei Technologies San Jose, CA
Research Intern 08/2022 — 09/2022

- Simulated RISC-V CPUs in Linux Full-System simulation mode using gem5.
- Cross-compiled binaries of SPEC CPU 2017 benchmarks to RISC-V for measuring the performance of CPU designs.
- Integrated SimPoint to checkpoint regions of interest (ROIs), accelerating simulations while preserving representativeness.

PUBLICATIONS

Amin Mamandipoor*, **Huy Dinh Tran***, Mohammad Alian, "SDT: Cutting Datacenter Tax Through Simultaneous Data-Delivery Threads," *IEEE CAL 2025*

* Equal contribution

PROJECTS

Simultaneous Multithreading (SMT) for gem5 2024 — 2025
Enabled SMT in gem5 O3CPU by restoring per-thread pipeline and interrupt handling, lifting the full-system single-thread restriction; implemented static and adaptive partitioning of microarchitectural resources across SMT threads.

Pipelined Execution for DPDK Userspace Networking Applications 2024
Implemented pipelined execution for DPDK-based userspace networking applications by decoupling NIC polling from application processing to enable parallelism; applied to DPDK l2fwd, Memcached and NGINX.

Building custom GPU power models with AccelWattch 2022 — 2023
Implemented a GPU power model using AccelWattch; profiled NVIDIA GTX 1050 Ti GPU to capture power usage and performance counters; simulated benchmarks in GPGPU-Sim to collect power model estimates.

SKILLS

Languages: C, C++, Python, Bash, LaTeX, MATLAB, Verilog, CUDA
Software & Tools: Git, gem5, Linux, Intel VTune, Intel CAT, DPDK, McPAT, Linux perf, GDB, Docker, OpenMP, GPGPU-Sim